Blase Londoño

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Education

University of California, San Diego, B.S. Mechanical Engineering — Expected Fall 2027

- GPA: 3.94/4.00

University of California, San Diego, M.S. Mechanical Engineering — Incoming Fall 2028

Experience

Mechanical Engineering Intern, Odin Dynamics, Inc — Sep 2026 – Present

- Designing UUV comms mast; antenna/RF mechanical integration, drag reduction, wake effects on thruster
- Owning full-vehicle wire harnessing; bulkhead integration, strain relief, connection selection, sealing & tolerancing

Mechanical Engineer, UCSD Yonder Dynamics — Oct 2025 – Present

- Designed and manufactured carbon fiber wheel hubs for Mars rover pneumatic wheels; Onshape modeling, wood-lathed mold, 3-ply vacuum bag wet layup, CNC post-processing; reduced hub weight by 50%
- Designed and manufactured TPU fin-ray compliant gripper for rover manipulator; FDM-printed ASA housing, heat-set inserts, rubber-laminated contact surface; fielded on 2026 competition rover

Mechanical Engineering Intern, NASA Ames Research Center — Jun 2026 – Aug 2026

- Designed instrumentation mounting hardware and wire routing in SOLIDWORKS for full-scale aerodynamic testing; produced DFM/GD&T drawings approved by aircraft mechanics and released for outside fabrication
- Diagnosed signal-clipping issue on Pacific Instruments strain-gauge boards; designed a low-cost KiCad adaptor PCB to correct signal reference from instrument input, recovered 2 channels/board & scaled across DAS
- Rapid-prototyped a modular vane set redesign in SOLIDWORKS for 1/50th-scale wind tunnel testing; 3D-printed airfoils, steel-reinforced to withstand 150kt airflow

Electrical Engineer, UCSD Yonder Dynamics — Oct 2024 – Oct 2025

- Designed 24V/60A rover power distribution PCB in Altium Designer; current-scaled trace widths, polygon pours for thermal management, per-fuse blowout indicator LEDs; reduced footprint by 20%

Research

Undergraduate Research Assistant, Gravish Bioinspired Robotics Lab — Apr 2026 – Present

- Implemented CAN bus network on Raspberry Pi to increase closed-loop control bandwidth for five-bar legged robot testbed; soldered/configured ODrive v3.6 controllers with Quantum MT5208 motors and absolute encoder feedback
- Diagnosed driver conflicts and specified an ODrive S1/CAN HAT upgrade to prevent future bottlenecks; rebuilt local software environment (VSCode, ODrivetool/Jupyter, SSH) to restore testbed capability

Projects

Thermoacoustic Refrigerator Project — Feb 2026 – May 2026

- Designed and manufactured thermoacoustic refrigerator (acoustic pressure heat pump) as project lead
- Modeled acoustic pressure in ANSYS Fluent, found resonator natural frequency for heat stack staging
- Achieved $4°$C temperature drop

Machining Seminar Kinematic Launcher — Mar 2026 – May 2026

- Manufactured spring-loaded kinematic launcher from supplied drawings; CNC-milled Delrin body/ball holder, CNC-lathed aluminum nozzle; manual-milled aluminum base plates and struts, manual-lathed firing pin
- Held ±0.003" across all mating features

Skills

Hardware: CAN Bus, Composite Layup, CNC Machining & Programming, DFM, Embedded Systems, GD&T, Laser Cutting, Manual Mill & Lathe, Raspberry Pi, Soldering, 3D Printing

Software: Altium Designer, ANSYS Fluent, Fusion 360, KiCad, MATLAB, ODrivetool, Onshape, Python, SOLIDWORKS